

SECTION 45 52 00

AEROSPACE X-RAY TESTING EQUIPMENT
06/25

PART 1 GENERAL

1.1 SCOPE

Provide labor, material, equipment and services necessary for, and reasonably incidental to, furnishing and installing aerospace x-ray testing equipment work indicated within the Instruments of Service. Incorporate related accessories and specialties to accomplish a complete and proper installation. Coordinate and schedule this work with the work of other trades to ultimately provide superior workmanship in the finished product.

Unless otherwise so qualified, engage a competent aerospace trade to perform this installation having five years of experience with work of a similar type, scope, complexity and scale.

1.2 SUBMITTALS

Government approval is required for submittals with a "G" or "S" classification. Submittals not having a "G" or "S" classification are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

TESTING EQUIPMENT; G

SD-03 Product Data

TESTING EQUIPMENT; G

Submit platform descriptions and manuals of programming, software, firmware and similar DIGITAL RECORDS that enable testing equipment system operations.

SD-05 Design Data

DELEGATED DESIGN; G

Submit a DELEGATED DESIGN demonstrating compliance

Furnish testing equipment systems. Technical specifications identify primary system components and basic functions. Assembly design responsibility is delegated to the Contractor and incorporates specific properties and equipment capabilities of selected products and manufacturers.

Testing equipment systems describe basic functions, exposures and characteristics. Furnish appurtenant equipment items, secondary components, connections, distribution, electronics and accessories to support conventional system operations. Compare suitability and adaptability of items cited with substrates and field conditions encountered. Account for supplementary items,

functional coordination and mounting options that optimize testing equipment use, operation and performance, and furnish these refinements to complement the design conditions scheduled.

The aerospace trade distributes power to the various systems serving the delegated design. The Instruments of Service only furnish power for the primary equipment. Design and provide supplemental distribution panels, cabinets, transformers, breakers, disconnects, wiring, cabling, electric metallic conduit, receptacles and supplemental mechanical equipment items. Rate supplemental items and devices for the climate and hazard category of conditions encountered.

Submit descriptions of each component type, color and finish of testing equipment and accessories proposed for use.

SD-06 Test Reports

ACCEPTANCE REPORTING ; G

SD-10 Operation and Maintenance Data

OPERATION AND MAINTENANCE; G

SD-11 Closeout Submittals

WARRANTY; G

1.3 DESIGN CRITERIA

1.3.1 TESTING EQUIPMENT

Aerospace x-ray testing equipment is custom engineered to facilitate nominal 33.7 foot by 2.72 foot, large blade inspections. Equipment is designed, manufactured, installed and supported by a single source. Completely assemble and test equipment at the point of manufacturing. Deliver for installation only after equipment passes factory acceptance testing. Controls are in a separate room within proximity of equipment. Acceptance testing is calibrated to Fleet Readiness Center East quality standards, which are conveyed to the aerospace trade for use during the submittal process. Furnish training for use and maintenance.

Aerospace x-ray testing equipment must comply with the requirements of NAVSEA manual S0420-AA-RAD-010, which includes timelines for design review of the equipment at 35% and 100% delegated design completion. Material and safety mechanism design and functionality requirements are also described therein. Applicable excerpts of NAVSEA manual S0420-AA-RAD-010 are conveyed to the aerospace trade prior to beginning the submittal process to ensure compliance.

1.3.1.1 Testing and Accepting Criteria

The inspection system shall be sufficient to provide a spatial resolution and contrast sensitivity to display a minimum of 3 of 4 contrast squares and a minimum of 4.0 LP/mm on the computer display when using the USN or USAF Computed Radiography Process Control Standard (Phantom).

Acceptance criteria and tolerances: The X-ray equipment shall operate for a minimum of five (5) consecutive eight (8) hour shifts without unplanned,

unexpected interruption. All pieces of equipment shall be of good quality and not physically damaged during or subsequent to their manufacture/assembly, shipment, or installation.

The contractor shall demonstrate that the X-ray equipment can process the parts without unplanned, unexpected interruption of operation. Unplanned, unexpected interruption shall be cause for repeat processing of inspection.

Static Test: The system shall provide a spatial resolution and contrast sensitivity to display a minimum of 3 of 4 contrast squares and a minimum of 4.0 LP/mm for line pair gauges A and B on the computer display when using the USN or USAF Computed Radiography Process Control Standard (Phantom).

3K Blades testing the contractor shall use the existing blade manipulator with the new mounting location to test for FOD and water or sealant on a H-53K main and tail rotor blade provided by the Government.

Safety Testing: The contractor shall perform tests to ensure that all safety features including fail-safe devices, interlock switches, electrical/electronic shielding and grounding, power loss protection, and other applicable items are installed and are functioning properly. In addition, the contractor shall fully demonstrate all lockout/tagout procedures. Audible noise levels shall be measured after installation for conformance with this specification.

1.3.1.2 Governments Documents Standards

Control of Hazardous Energy (Lock-out/Tag-out). A written procedure for locking out all energy sources shall be provided by the contractor in accordance with 29 CFR 1910.147, reference CDRL A002 Lockout and Tagout Procedure. The contractor shall submit draft copies to the Government in accordance with the contractor submitted and Government approved Plan of Action and Milestones and final copies are due at start of equipment installation. The contractor shall provide an electronic copy of the procedure in Microsoft Word to the Government as well as include the procedure in the technical manuals.

Plan of Action and Milestones - The contractor shall provide a Plan of Action and Milestones (POA&M) to identify contractor performance dates of all major events from contract award to completion of all contract requirements, reference CDRL A007 Plan of Action & Milestones. The initial POA&M shall be delivered at the time of proposal. The milestones shall be prepared using a recognized project management technique, preferably Microsoft Project). The Government and the contractor shall review the report on a monthly basis. The contractor shall notify the Government prior to making any changes to the POA&M. All changes to the POA&M must be approved in advance by the Government. The POA&M shall detail such items as system design concept, system development, software design concept, software development, engineering, procurement of major hardware, assembly, operational debug, factory acceptance, shipment, installation, training, and final acceptance test.

Acceptance Test Plan - The contractor shall prepare an acceptance test plan for the Blades, reference CDRL A009 Acceptance Test Plan. The plan shall be a step-by-step written instructions on how to verify that the equipment can inspect blades in accordance with the following documents:

A1-861PA-150-600 - Main Rotor Blade Assembly

A1-489PB-150-100 - Rotary Rudder Blade Assembly

NAVAIR 01-1A-16-2/T0 33B-1-2 - Nondestructive Inspection General
Procedures and Process Controls
ASTM E1255 Standard Practice for Radioscopy

1.3.1.3 Quality Assurance

Responsibility for Inspection and Compliance - The contractor shall be responsible for the performance of all inspection requirements specified herein. The Government reserves the right to perform any inspections set forth in the purchase description where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Contract Quality Assurance Inspection (Origin) - Contract quality assurance inspection, reference CDRL A013 Quality Assurance Provisions, shall be performed at the contractor's facility in accordance with the contractor submitted and Government approved POA&M prior to being provided for acceptance under the contract.

Final Inspection/Acceptance Test (Destination) - Final inspection and acceptance test shall be performed on the equipment in accordance with the contractor submitted and Government approved POA&M to ensure conformance with this specification, reference CDRL A009 Acceptance Test Plan. The acceptance test shall be performed only after the equipment is installed at its final location.
Inspection and Acceptance

Preliminary Inspection at Origin with Final Inspection and Acceptance at Destination: Preliminary inspection of the machine shall be at origin by a Government representative with final inspection and acceptance at destination by a Government representative upon completion of installation at the Government's facility by the contractor.

1.3.1.4 Power

Vault and Manipulator Equipment: 480 Volt, 3 Phase, 60 Hertz, 50 Amp
X-ray Equipment: 208 Volt, 1 Phase, 60 Hertz, 40 Amp
Computer Hardware: 120 Volt, 15 Amp

1.4 WARRANTY

The contractor must provide an initial warranty of 12 months, to commence at the completion of final acceptance. During the warranty period, supplier shall assume responsibility for any and all defects in materials, equipment, workmanship and performance. Under this warranty, the labor necessary to repair the equipment and all associated travel expenses will be provided by the vendor at no extra cost to the Government.

PART 2 PRODUCTS

2.1 MATERIALS

2.2 EQUIPMENT

Provide material as indicated in the Aerospace X-ray Testing Equipment Schedule below.

Furnish TESTING EQUIPMENT indicated.

X-ray vaults are constructed of shielded modular panels and accommodate digital, real time imaging functions. Vaults comply with 21 CFR 1020.40 and include electrical and lighting distribution. Equipment is 160 kV source, cooler and controller. Furnish a safety interlock system with warning lights and signals accessed via doors within the shielded enclosure. Equipment features an "X-ray ON" lighted signage when operating.

Furnish computer hardware and accessories capable of operating equipment software provided.

2.3 ACCESSORIES

Furnish fasteners, accessories and options as recommended by the manufacturer for the specific applications and substrates encountered within the work.

PART 3 EXECUTION

3.1 CONTRACT COORDINATION

The General Contractor coordinates with the aerospace and electrical trades to locate concealed conduit within walls and junction boxes above ceilings. The electrical trade installs conduit raceway and provides the line side feeder, junction box, cover and conductors of sufficient length. The aerospace trade provides materials to make load side connections and complete wiring to equipment being powered.

3.2 INSTALLATION

3.2.1 DELEGATED DESIGN

Aerospace x-ray testing equipment requires an 8 inch slab reinforced with No. 4 bars at 1 foot, 0 inches each way.

Prepare a DELEGATED DESIGN for aerospace x-ray testing equipment assemblies.

The manufacturer performs equipment start up and testing. Test and adjust controls and safeties. Cycle through the full range of operational functions. Replace malfunctioning controls and components. Check the general mechanical operation of motors. Perform preventative maintenance, including but not necessarily limited to, lubricating oil bearings; flush, clean and lubricate grease bearings; disconnect couplings and check electric motor rotation. Correct equipment malfunctions and make adjustments to provide smoothly operating systems. Summarize testing procedures, performance and manufacturer installation acceptance in reporting.

Provide programming, software, firmware and similar DIGITAL RECORDS. Furnish software authentication codes, license numbers, passwords and similar information necessary to recover and reinstall operating system platforms.

The vaults need to comply with 21 CFR1020.40 and NAVSEA S0420-AA-RAD-010 Section 11.

Provide **ACCEPTANCE REPORTING** for recordkeeping. Preliminary Inspection at Origin with Final Inspection and Acceptance at Destination. Preliminary inspection of the machine will be at origin by a Government representative with final inspection and acceptance at destination by a Government representative upon completion of installation at the Government's facility by the contractor.

The aerospace trade to conduct the following training:

Operator training - four (4) X-ray operators, for a minimum of 2 days of 8 hours

Mechanical and Electronics/Electrical Maintenance training - two (2) mechanical and two (2) electronics/electrical maintenance for a minimum of 2 days of 8 hours.

Engineering training: a minimum of 2 days of 8 hours.

3.3 OPERATION AND MAINTENANCE

Submit the recommended maintenance practices for each type of installation provided.

3.4 AEROSPACE X-RAY TESTING EQUIPMENT SCHEDULE

XTE-1 Aerospace X-ray Testing Equipment

1. Exterior dimensions: Nominal 46 feet long by 14 feet wide by 11 feet tall
2. Interior dimensions: Nominal 45 feet long by 12 feet wide by 10 feet tall
3. Door type: Nominal 6 foot width and 8 foot tall, biparting motorized slider
4. X-ray scanning: Gantry mounted x-ray source and detector manipulator with servo control motion
5. Blade manipulator: not rotational
The detector:
 - a. raise and lower the tube and detector (up and down)
 - b. move the tube and detector along the span of the blade (left and right)
 - c. move the tube and detector towards and away from the blade (in and out)
6. Control software: Teach and learn motion control
7. Internal bridge crane: 1/2 ton (1,000 lbs), for loading blades from transport cart to blade manipulator
The internal crane moves horizontal and vertically and needs to load the blade from transport cart to blade manipulator.
8. X-ray characteristics: Mark X system with MXR 160-HP11 having 0.4 to 1.0 focal spots at 800 to 1,800W, 160 kV source, cooler and controller
9. Detector characteristics: 1515DX detector with 28 FPS, 3.94 lp/mm limiting resolution, 14.6cm x 14.6cm image size, 127 um^2 pixel pitch, high speed scanning capability to detect foreign objects and water intrusion
The system should be able to perform a complete scan of a 53K blade within 1.5 hours. Will require a frame rate of 28 fps or

better.

10. Digital imaging panel: Enables real time scan ability with video recording and single image capture to display and save 2D stationary high resolution radiographic imagery for real time inspections and static inspections

Real Time: The system must detect water intrusion, crushed core, voids, foreign object inclusion of rotor blades in real-time mode.

Static: The system must provide a spatial resolution and contrast sensitivity sufficient to display a minimum of 3 of 4 contrast squares and a minimum of 4.0 LP/mm for line pair gauges A and B on the computer display when using the USN or USAF Computed Radiography Process Control Standard (Phantom).

11. Imaging software: Image acquisition and analysis
12. Safety interlock system: Complying with Fleet Readiness Center East requirements
13. Operator accessory: Control station based on control room dimensions
14. Electrical: NEMA ICS-6, NEMA 250 Type 12
15. Control room: Nominal 10 feet long by 12 feet wide
16. Axes:
- a. Move the source and detector along the length of the blade
 - b. Move the source and detector along the height of the blade
 - c. Rotate the blade
 - d. Move the source towards and away from the blade independent of detector (change source-to-object distance)
 - e. Move the detector towards and away from the blade independent of the source (change object-to-detector distance)
 - f. Angulate the synchronized source and detector along x-axis a minimum of 30 degrees
 - g. Capability to move the source and detector independently along each axis.
 - h. The system shall be able to angulate the tube and detector (simultaneously) a minimum of + / - 30 degrees (45 degrees preferred) from perpendicular to the blade
 - i. The system shall have a range of motion for each axis must be sufficient to cover 35' span by 3' chord rotor blade

CH-1 Aerospace X-ray Testing Equipment Computer Hardware and Accessories

- a. Dell Precision 7865 Tower (or similar)
- b. Windows 11 Pro English 23H2
- c. AMD Ryzen™ Threadripper™ PRO 5000 WX-Series
- d. 64 GB, 1x32 GB NECC 4800MHz DDR5 CAMM Module, minimum
- e. 4TB, Gen 4 PCIe x4 NVMe, Solid State Drive
- f. 1 each, power cord 110 Volt
- g. 4 each, USB ports
- h. Two Dell UltraSharp 27 InfinityEdge with Dell MDS19 Dual Monitor Stand
- i. 1 each, Dell Smartcard KB813 and corded mouse

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